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Multimedia Tools and Applications
An International Journal

ISSN 1380-7501

Multimed Tools Appl
DOI 10.1007/s11042-019-07929-y

ONLINE FIRST

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Multimedia Tools and Applications
https://doi.org/10.1007/s11042-019-07929-y

Automatic image annotation via category labels

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Received: 25 September 2018 / Revised: 13 May 2019 / Accepted: 21 June 2019 /
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Abstract
Automatic image annotation aims to assign relevant keywords to images and has become a research focus. Although many techniques have been proposed to solve this problem in the last decade, giving promissing performance on standard datasets, we propose a novel automatic image annotation technique in this paper. Our method uses a label transfer mechanism to automatically recommend those promising tags to each image by using the category information of images. As image representation is one of the key technique in image annotation, we use sparse coding based spatial pyramid matching and deep convolutional neural networks to model image features. And metric learning technique is further used to combine these features to achieve more effective image representation in this paper. Experimental results illustrate that the proposed method get similar or better results than the state-of-the-art methods on three standard image datasets.

Keywords Automatic image annotation · Image understanding · Deep learning · Sparse coding

1 Introduction

Users and systems can better understand the semantic content of media aided by annotation which is also the basis of many text based image processing techniques, such as text based

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image retrieval [20], text aided image classification [35] etc. However, annotating images manually requires time and effort, and it is difficult for users to annotate all relevant tags for each image. Thus automatic image annotation emerged and has attracted lots of attention.

Given an input image, the goal of automatic image annotation is to assign a few relevant text words to the image which can mainly reflect its semantical content. It will allow one to further exploit the fast indexing and retrieval architecture of web image search engines for improved image search by utilizing image content to assign a richer, more relevant keywords. Image annotation has been a hot research topic and a number of techniques have been proposed [1, 4–7, 11, 22, 23, 28, 33, 34]. Most of these methods define a parametric or non-parametric model to capture the relationship between image features and keywords.

Even though some of these traditional techniques have shown impressive results, they considered all the images in the training set are relevant with the testing image. Hence, to get the appropriate tags, we need compare the testing image with each image in the training image set and then propagate the tags of training images to the testing image. Obviously, this kind of approaches are time-consuming. We have also published a solution to automatic image annotation in ICMLC 2012 [44]. In this paper, we extend and improve our previous work. We proposed a novel method to address automatic annotate images based on image category tags. Our method utilizes the category information which can be used to filter out those content irrelevant images and the remaining of relatively clean images are used for tag propagation. On the other hand, image representation is one of the key technique in image annotation and it will directly affect the performance of annotation. In this paper we represent images using sparse coding based spatial pyramid matching to model image's SIFT features. And we also use the pre-trained deep convolutional neural networks (CNN) as feature extractors, which has become a popular technique in various computer vision tasks [29, 38]. Furthermore, category information is also used to learn a more effective image distance metric, which combines the above two kinds of features. Experimental results show that the keywords recommended by our model can effectively reflect the image content.

The rest of this paper is organized as follows: In Section 2, we review some classical image annotation systems. We highlight the main contributions of our work by analyzing the differences between our work and these works. We give an overview of our model in Section 3. Then we demonstrate image representation which is a key technique in our method in Section 4 before discussing the keyword propagation in Section 5. The evaluation of our method and experimental setup will be given in Section 6. And we conclude our work in the end.

2 Related work

A large number of image annotation techniques have been proposed in the last decades. Parametric or non-parametric models are used to build the relationship between images and labels. Here we roughly review the development course of automatic image annotation task.

Early works in this field, such as Cross Media Relevance Models(CMRM) [10], Continuous Relevance Model(CRM) [15], and Multiple Bernoulli Relevance Model(MBRM) [4], assume different, non-parametric density representations of the joint word-image space. CMRM simply approximates the probability of observing a set of blobs and words in a given image. CRM argued that the process of quantization from continuous image features into discrete blobs, as the approach used by the CMRM model, using continuous

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probability density functions to estimate the probability of observing a region given an image. Feng et al. modified the CRM model such that the probability of observing labels given an image is modeled as a multiple-Bernoulli distribution.

Recent research efforts have focused on building a high effective, even real time annotation recommendation system [25, 27]. In a image tag recommendation system, several tags are randomly initialized beforehand for each image. Based on the given tag and visual feature, some tags are recommended to this image. Li and Wang [18] developed a real-time ALIPR image search engine which uses multiresolution 2D HiddenMarkov Models to model concepts determined by a training set. While this method successfully infers higher level semantic concepts based on global features, identification of more specific categories and objects remains a challenge.

Most recently, deep learning algorithms [30, 37], which have shown its great power in single-label image classification [13], are used to automatically learning image features extraction and showed significant performance gain in large image datasets [5, 11–13]. And deep learning has also been adopted to solve image auto-annotation task. Kiros et al. [12] use convolutional neural network (CNN) [13] to build a hierarchical model for learning image representations from the pixel level and then added an autoencoder which has only one hidden layer to learn binary codes for annotations. Literature [32] proposes a method which directly used a pre-trained network to extract visual features and experimental results show the CCA based model gained remarkable benefit from these deep learning features. Furthermore, literature [36] proposes a CNN-RNN framework to learn a joint image-label embedding to exploit the label dependency and image-label relevance. However, these models still can not achieve even comparable results with the models combing CCA or KCCA embedding and nearest-neighbor based label propagation models.

Even though promising results have been reported by many sophisticated annotation techniques, our work proposed another feasible approach to annotate images. The contributions of this paper can be summarized as follows: (i) Our method uses category information to aid image annotation, which helps filter out content irrelevant images and improve the annotation performance and lower down the computational complexity. (ii) Sparse coding based spatial pyramid matching and deep convolutional neural networks are used to model image's features. And metric learning technique is adopted to combine these two kinds of features together, which significantly enhances the annotation performance.

3 Model overview

In this paper, we propose a novel automatic image annotation model as shown in Fig. 1. Our method mainly contains four modules: (i) Image feature extraction: traditional hand-crafted visual features and neural network based visual features will be extracted to represent the image. (ii) Image distance metric learning: we learn a image distance metric in a data-driven manner. This metric will be used to measure the similarity of the input image and the relevant images. (iii) Noisy image filtering: we first generate category labels from the images' category information for the input image and then noisy images are filtered out by using these labels as the keyword. (iv) Tag propagation: tags corresponding to the relevant images are propagated to the input image based on the learned distance and statistical relationship. Note that, in our model, the input image is only compared with the relevant images which is a subset of the large training set while traditional annotation approaches compare the input image with all the training images.

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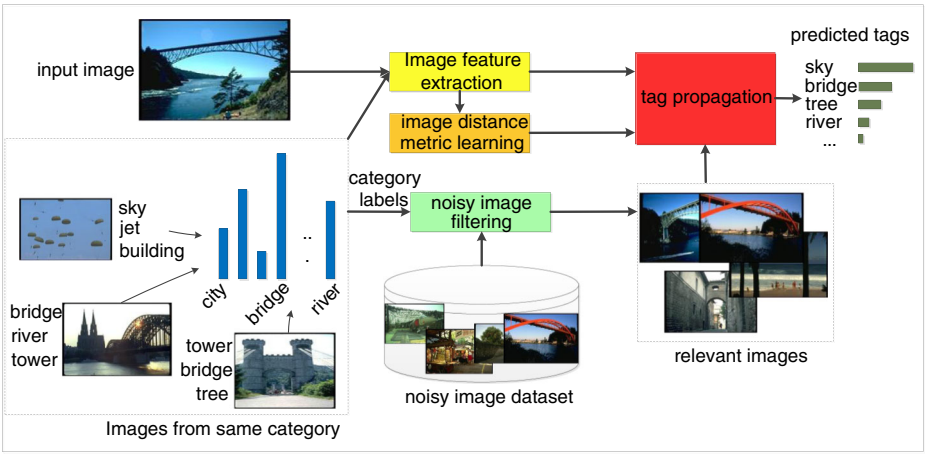

Fig. 1 The framework of our automatic image annotation model. We select images which are in the same category as the input image and get the label distribution of these images to generate category labels. After expanding these category labels, noisy images are filtered by keywords matching. Meanwhile, image features are extracted and a metric learning technique is used to learn a image distance metric. Finally, the labels of relevant images will propagate to the input image according their visual similarity

4 Image representation

Image representation is one of the key techniques in our proposed annotation system. This section introduces our image representation based on hand-crafted visual features and neural network based feature.

4.1 Representation based on hand-crafted features

Traditional image representation consists of two major approaches: bag-of-features(BoF) and spatial pyramid matching. The BoF method represents an image as a histogram of its local features. It is especially robust against spatial translations of features, and demonstrates decent performance in whole-image categorization tasks. However, the BoF method disregards the spatial information, hence it is incapable of capturing shapes or locating an object. Of the many extensions of the BoF method, the most successful results were reported by using Spatial Pyramid Matching(SPM). The SPM method partitions the image into increasingly finer spatial sub-regions and computes histograms of local features from each sub-region. Typically, $2^l \times 2^l$ subregions, l =0, 1, 2 are used. The resulted "spatial pyramid" is a computationally efficient extension of the orderless BoF representation, and has shown promising performance on image classification tasks. For more information about SPM, please see [16]. Although the traditional SPM approach works well for image classification, people argue that the hard vector quantization in codebook generation is not proper and Yang *et al*. use sparse coding to obtain nonlinear codes [40].

Sparse coding for code book generation k-means clustering is widely used for the codebook generation in BoW model. In k-means method, the data matrix of local feature space $X = [x_1, x_2, ..., x_N]$ ($x_i \in \mathbb{R}^{d\times 1}$ is the SIFT feature of a local key point.) is partitioned into k clusterings $S = [S_1, S_2, ..., S_k]$, and the corresponding centers $U = [u_1, u_2, ..., u_k]$

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$\in \mathbb{R}^{dxk}$ form the codebook. In k-means clustering, each local feature is assigned to one clustering center only, and the weight contributing to that center is 1.

However, this constraint is too strict, especially for those points located at the boundary of several clusters. To this end, the hard constraint $Card(v_i) = 1$ on v_i is usually relaxed for the soft vector quantization. Moreover, to avoid each feature to be assigned to many clusters, the sparse constraint on the weight vector v_i is usually added to the objective. Then we arrive at the optimization problem of sparse coding:

$$\begin{aligned} &\min_{U,V} \sum_{i=1}^{N} ||x_i - Uv_i||^2 + \lambda ||v_i||_1 \\ &subjectto : |u_j| \leq 1, \forall j = 1, ..., k. \end{aligned} \tag{1}$$

The optimization problem (1) is convex in V (with U fixed) and convex in U (with V fixed), but not in both simultaneously. Lee et al. designed a efficient sparse coding algorithms[1] to solve the above optimization problem [17].

Spatial pyramid matching based image representation After sparse coding, we can learn both the codebook, in which each basis vector represents one basic local patch pattern, and hundreds of thousands of sparse codes. Each entry of certain sparse code represents the response of the patch to the corresponding basic pattern in the codebook. To preserve the spatial information, SPM is used. Following the work of Yang et al. [40], maximum pooling based image representation is used. Suppose one image region has n local features, and the codebook size is K. The sparse codes for these local features are $[v_1, v_2, ..., v_n]$. After maximum pooling, each image will be represented by a K dimensional vector $f_{SPM} = [y_1, y_2, ..., y_K]$, and the lth entry is the largest response to the lth basis vector of the all the sparse codes in selected region (v_{nl} is the lth entry v_n):

$$y_l = max\{|v_{1l}|, |v_{2l}|, ..., |v_{nl}|\} \tag{2}$$

4.2 Representation based on neural network

Recently, deep neural networks [13] have proven their great power in many artificial intelligence tasks, such as computer vision [2, 26, 31] and natural language processing [8, 14, 39]. And in many works, the pre-trained deep neural networks on large image datasets can be used as image feature extractors, and have been successfully applied in cross-modal retrieval [43], image annotation [12, 45], visual question answering [41, 42] and other tasks. Follow this idea, we resize the images and extract image features using the pre-trained ResNet [9]. In practice, we extracted the 2048 dimensional activations of the layer right before the last fully-connected layer, denoted as f_{CNN}.

4.3 Image distance metric learning

Now we have got two kinds of visual features for each image. As we have discussed in Section 1, image similarity metric plays an import role in automatic image annotation. A proper image distance metric should be able to reflect the semantical relationship between

[1] The source code of this algorithm can be downloaded from http://www.eecs.umich.edu/~honglak/softwares/nips06-sparsecoding.htm

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images. In this paper, we use metric learning technique to learn such a proper image distance metric. Given two images I and J, we define their distance as follows:

$$\begin{aligned} d(I,J) = \alpha \|relu(W(f_{SPM}(I)) - relu(W(f_{SPM}(J)))\|_2 \\ +(1-\alpha)\|relu(V(f_{CNN}(I))) - relu(V(f_{CNN}(J)))\|_2 \end{aligned} \tag{3}$$

where $W \in R^{D\times|f_{SPM}|}$ and $V \in R^{D\times|f_{CNN}|}$ are the matrix need to be learned. $\|\cdot\|_2$ means $L2$ norm and the hyper-parameter α controls the weights of hand-crafted features and neural network based features. This distance definition can be understood like that we first linearly project the SPM features and CNN features into a higher dimensional space by multiply by matrix W and V, then $relu$ activation function is used to introduce nonlinearity. Finally we combine them together using a weighted sum manner.

To learn the parameters in our metric learning model, we use maximum margin loss function to train our model:

$$Loss = max(0, m - (u^+ - u^-)) \tag{4}$$

where m is the margin. This objective function maximizes the mean similarity score u^+ between images from the same category while minimizes the mean similarity score u^- between images from different categories.

5 Image annotation

Noisy image filtering Images in the large training image set are very noisy: lots of them are content irrelevant to the image to be annotated. Directly using these noisy images may worsen the performance of annotation and increase computational complexity, so it is necessary to get rid of irrelevant images. Therefore, we first generate category labels for the input image. As shown in Fig. 1, the labels of all the images from the same category as the input image are selected and the label frequency is calculated as $[p_{w_1}, p_{W_2}, ...p_{w_N}]$, where

$$p_{w_i} = \frac{1}{M}\sum_{I_j \in \mathbb{C}} h(I_j, w_i) \tag{5}$$

N and M is the vocabulary size of the training set and the total number of labels existing in all the images in category $\mathbb{C}$. $h(I, w)$ is equal to one if image I has label w, zero otherwise. Then we use the top B frequent labels as keywords to filter these noisy images and get relevant images which is a subset of the large training set.

Tag propagation Tag propagation module tries to propagate the relevant images' labels to input image based the joint probability of observing an image with possible annotation keywords. Given a training image set $\mathcal{T}$, each image $J \in \mathcal{T}$ is assumed to be generated from:

$$P(J, w) = P(w|J)P(J) \tag{6}$$

where w is a annotation attached to image J. The above process that generates J is based on two distinct probability distributions: $P(w|J)$ is the conditional probability of generating w given J, and $P(J)$ is the probability of selecting the underlying model of image J, typically considered as uniform distribution.

Given a test image $I \notin \mathcal{T}$, we assume that I comes from the same process that generates each training image $J \in \mathcal{T}$. However, without the annotation of I, we do not know which

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process that is. Hence, we compute an expectation over all images $J \in \mathcal{T}$ and the joint probability of a set of keywords W and a test image I is given by:

$$\begin{aligned} P(I, w) &= \sum_{J \in \mathcal{T}} P(I, w|J)P(J) \\ &= \sum_{J \in \mathcal{T}} P(w|J)P(I|J)P(J) \\ &= \sum_{J \in \mathcal{T}} P(w|J)P(I, J) \end{aligned} \tag{7}$$

where the test image I is assumed to be independent of W given a training image J. Although any kind of distribution can be used to model annotation keywords (i.e. $P(w|J)$) in the above relevance model, we follow the Laverenko's method [15]:

$$P(w|J) = \frac{\mu p_w + N_{w,J}}{\mu + \sum_{w'} N_{w',J}} \tag{8}$$

where $N_{w',J}$ is the number of times the keyword w appears in the annotations of training image J. P_w is the frequency of keyword w in the whole training set and μ is a parameter which determines the degree of interpolation, with a larger value of μ given more precedent to the background probability over the image annotation word frequencies. We fix it to be 0.01 in all of our experiments.

Finally, we need decide how to calculate the conditional distribution $P(I|J)$ which actually measures the similarity between testing image I and training image J:

$$P(I|J) = \frac{s(I, J)}{\sum_{j=1}^{R} s(I, J_j)} \tag{9}$$

where R is the number of relevant images, and

$$s(I, J) = \frac{e^{-d(I,J)}}{\sum_{j=1}^{R} e^{-d(I,J_j)}} \tag{10}$$

6 Experiment

6.1 Dataset and performance measures

Our experiments examined the performance of the proposed image annotation method on three widely used standard image datasets.

Corel5k is a basic comparative dataset for recent research works on image annotation [3, 4, 10, 15]. There are 5,000 images from 50 Stock Photo CDs in this dataset. Each CD includes 100 images on the same topic. Each image is annotated with 1 to 5 keywords and totally there are 374 keywords in the dataset. Out of the 5,000 images, 4,500 images are used for model training and the other 500 images are used for testing.

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MSRC image database[2] contains 591 images annotated with 23 keywords. In particular, each image is annotated with 1-7 keywords. the images are proximately of the size 320x240 pixels. We use 15 images for training for each class, and the other for testing.

NUS-WIDE This dataset has 269648 images collected from Flickr. Most of these images are manually annotated with several labels, leading to 81 labels in total for the whole dataset. Following [5], we have 209347 images after discarding images with no labels and we select a subset of 150k images for training and the others are left for testing. 15k images in the training set are selected as validation set.

There are several parameters to evaluate the performance of automatic image annotation algorithms. We use the most popular measures including:

Word Recall is the number of images correctly annotated with a given word, divided by the number of images that have that word in the human annotation. This metric measures the completeness in annotating images with word w:

$$r_w = \frac{c_w}{e_w} \tag{11}$$

Here c_w is the number of correct images annotated with word w and e_w the number of images annotated with w in the ground truth.

Word Precision is the number of correctly annotated images divided by the total number of images annotated with that particular word(correctly or not). This metric measures the accuracy in annotating images with word w:

$$p_w = \frac{c_w}{r_w} \tag{12}$$

Here r_w is the number of images the system has annotated with word w.

Number of words with recall greater than zero ($N+$): This metric seeks to measure the ability of the system to label images with rare keywords which are hard to annotate due to the small number of positive instance in the training set. This metric is also important as it is possible to achieve high precision and recall values by performing very well on a small selection of common words.

Typically the single word recall and precision are averaged over all words that exit in the testing dataset. The mean per word precision is given by:

$$\overline{P_n} = \frac{1}{n}\sum_{w=1}^{n} p_w \tag{13}$$

With the mean per word recall expressed as:

$$\overline{R_n} = \frac{1}{n}\sum_{w=1}^{n} r_w \tag{14}$$

6.2 Comparison with state-of-the-arts

In this experiment, we compare the performance of our model on the three datasets, with some classical models and recently published state-of-the-art models.

For *Corel5k* dataset, we compare our results with CRM [15], MBRM [4], SMK+GRM [21], TagProp [6], 2PKNN [33], SKL-CRM [24], and CCA-KNN [32]. The results are

[2]MSRC images can be downloaded from http://research.microsoft.com/en-us/projects/objectclassrecognition/

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Table 1 Results on Corel5k

	$N+$	$\overline{R_n}$	$\overline{P_n}$
CRM	107	.19	.16
MBRM	122	.25	.24
SMK+GRM	143	.33	.30
TagProp	160	.42	.33
2PKNN	191	.46	.44
SKL-CRM	184	.46	.39
CCA-KNN	192	.51	.39
Our model	228	.68	.60

shown in Table 1. The results of our experiments demonstrate that our model outperforms all these state-of-the-art methods we have reviewed by a large margin. Our method achieves P, R and $N+$ scores of 0.60, 0.68, and 228 on the $Corel5k$ dataset. On $MSRC$ dataset, our model also significantly improves the annotation performance, compared with MBRM [4] and SMK+GRM [21], as shown in Table 2. Here, we should point out that, being different from the state-of-the-art models, our model utilizes the category information, that means our model get more prior knowledge of the input images.

To demonstrate the scalability of our model, we also test it on a popular large scale dataset *NUS-WIDE* which has 269648 images. The performance of our model is compared with five recently proposed models including MLRank [19], CNN+WARP [5], CNN+KNN [11], CNN-RNN [36] and Neighbors-meta [11], as shown in Table 3. MLRank treats automatic image annotation as a label ranking problem and proposes a complex ranking model to solve it. CNN+WARP model trains convolutional neural networks from scratch using WARP loss to achieve end-to-end image annotation, obtaining an inferior performance with respect to the models using pre-trained CNN features, such as CNN+KNN model. Wang et al. attempted to learn joint image-label embedding using CNN-RNN framework [36], getting unsatisfactory performance. Our model gets similar results as the state-of-the-art model proposed by Johnson et al. [11], which also exploits image metadata including category information.

6.3 Parameters analysis

In this experiment, we evaluate the impact of the parameters on the annotation performance. Two important parameters including the first dimension size of matrix W and V, denoted by D, and α (alpha) who controls the weights of SPM features generated distance and CNN features generated distance, are tested. We set D to be 1024, 2048, and 4096 in turn. Meanwhile, the parameter α (alpha) in (3) varies from 0 to 1 with step size 0.1. Thus we get 33 pairs of $\{D - \alpha\}$ and we record the annotation performances of all these parameter combinations. For ease of presentation and comparison, we use another common performance

Table 2 Results on MSRC

	$N+$	$\overline{R_n}$	$\overline{P_n}$
MBRM	22	.52	.42
SMK+GRM	22	.62	.61
Our model	22	.75	.72

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Table 3 Results on NUS-WIDE

	$N+$	$\overline{R_n}$	$\overline{P_n}$
MLRank	–	.28	.24
CNN+WARP	78	.31	.32
CNN+KNN	–	.31	.44
CNN-RNN	–	.30	.41
Neighbors-meta	–	.57	.55
Our model	80	.54	.51

$F1 = 2 * \overline{P_n} * \overline{R_n}/(\overline{P_n} + \overline{R_n})$, which is the trade-off between mean word precision and recall. The results are shown in Fig. 2. It is obvious that highest performance is obtained when $D = 2048$ and $\alpha = 0.4$ for $Corel5k$ dataset. On $MSRC$, $D = 1024$ and $\alpha = 0.5$ is the optimal choice, while D and α should be 4096 and 0.3 respectively on *NUS-WIDE* dataset. Setting a larger value to D means giving higher capacity to the image distance metric learning model, and higher model capacity need more samples to train. Therefore, the largest dataset *NUS-WIDE* needs largest D and too large value of D is not proper for small dataset, since it may cause overfitting. In addition, we observe that the performances on all of the three datasets drop rapidly when we set α to be 0 or 1, which means we only use the SPM features or the CNN features. In other words, combining these two kinds of features can significantly improves the annotation performance.

6.4 The impact of distance metric learning

In this paper, we use two kinds of visual features to represent image, including spatial pyramid matching based image representation and CNN based representation. And image distance metric, which plays an important role in automatic image annotation task, should

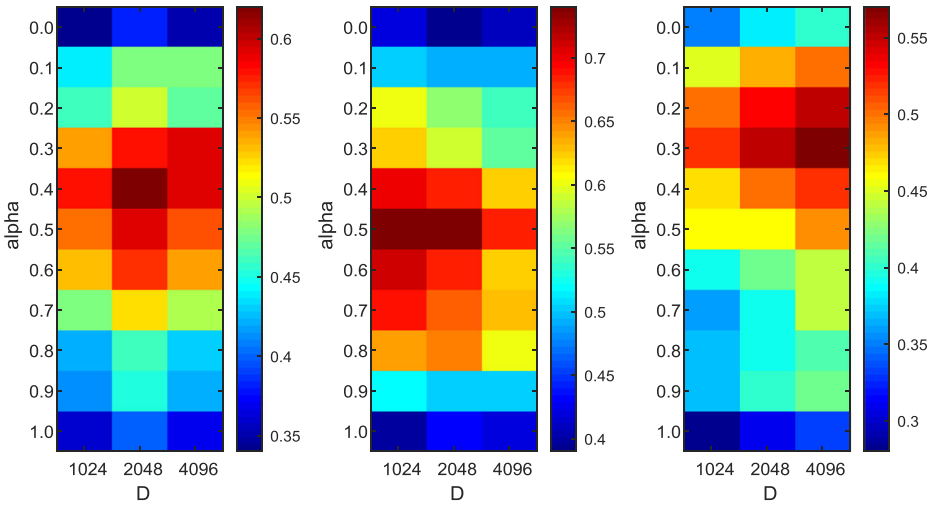

Fig. 2 The results of parameters analysis. We set the first the dimension of matrix W and V in image distance metric learning to be 1024, 2048, and 4096. Meanwhile, the parameter α (alpha) in (3) varies from 0 to 1 with step size 0.1

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Table 4 Comparison of performance of our baseline model and full model

Dataset	Baseline model			Full model		
	$N+$	$\overline{R_n}$	$\overline{P_n}$	$N+$	$\overline{R_n}$	$\overline{P_n}$
Corel 5k	190	.50	.45	228	.68	.60
MSRC	22	.59	.55	22	.75	.72
NUS-WIDE	78	.45	.44	80	.54	.51

be able to reflect the semantical similarities between images. We argue that proper image distance should consider all of these image representations. Thus, we propose a metric learning algorithm to evaluate image distance by combining hand-crafted features and features extracted using deep network (CNN). In this experiment, we conduct ablation tests to prove the effectiveness of our proposed distance metric learning approach. The variant versions of our model include:

- Baseline model: In this baseline model, given two images I and J, their distance is calculated using:

$$d(I, J) = ||f_{SPM}(I) - f_{SPM}(J)||_2 + ||f_{CNN}(I) - f_{CNN}(J)||_2 \tag{15}$$

- Full model: In this model, the image distance is calculated using (3).

Table 4 shows the result. It is observed that our full model which has adopted image distance metric learning module significantly outperforms the baseline model. These experimental results prove that the learned distance metric helps the testing image find more relevant training images and finally boost the annotation performance.

7 Conclusions

In this paper, we propose a novel automatic image annotation model. Being different from traditional approaches, we utilize the category information of the images to achieve better performance. These metadata is not only used to filter out noisy images, but also helps us learn a better image distance metric. Our experimental results on three standard datasets including $Corel5K$, $MSRC$, and large-scale *NUS-WIDE* demonstrate that our method makes remarkable improvements on all of these datasets, compared with existing state-of-the-art methods.

Acknowledgements This work is supported by the Natural Science Foundation of China (No. 61572162) and the Zhejiang Provincial Key Science and Technology Project Foundation (No. 2018C01012).

Funding This work is supported by the Natural Science Foundation of China (No. 61572162) and the Zhejiang Provincial Key Science and Technology Project Foundation (No. 2018C01012).

Compliance with Ethical Standards

Conflict of interests The authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

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